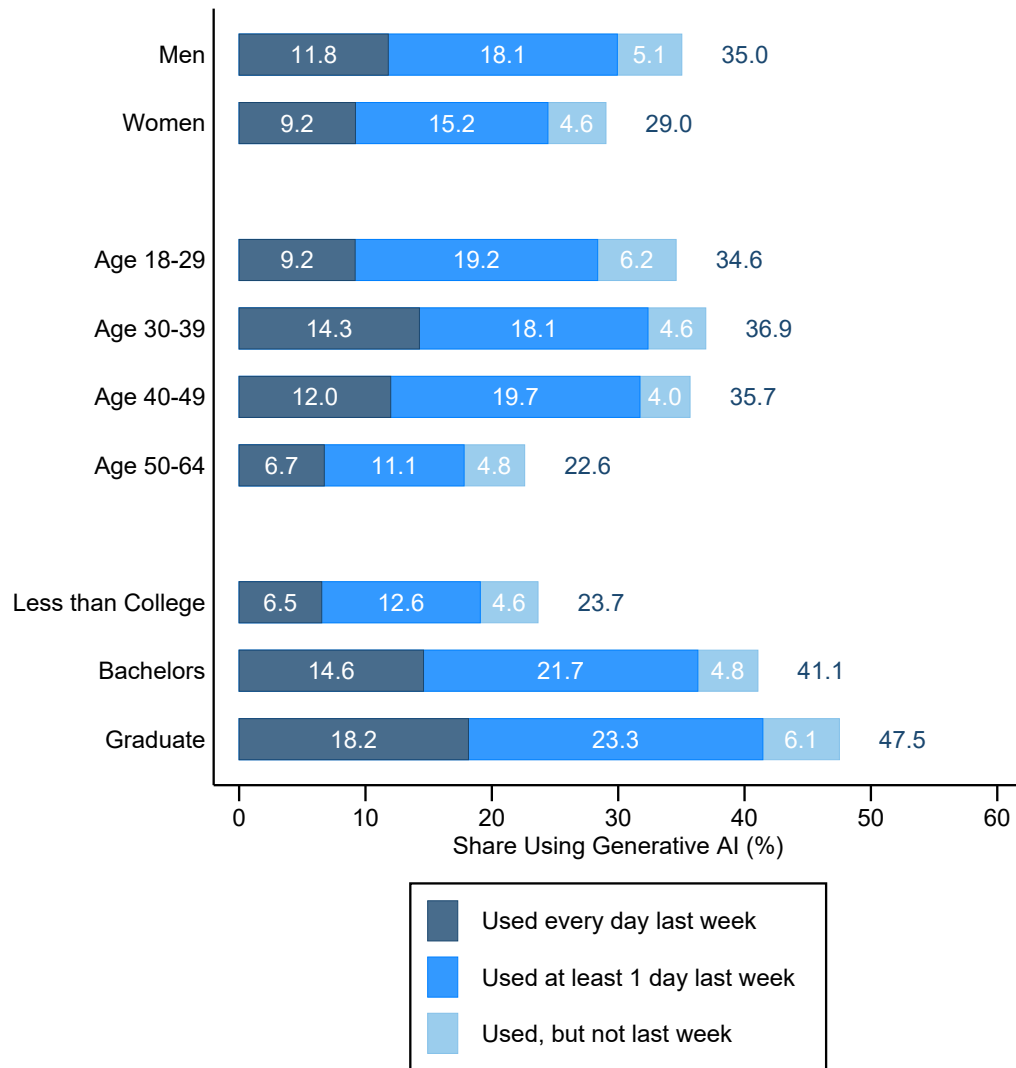
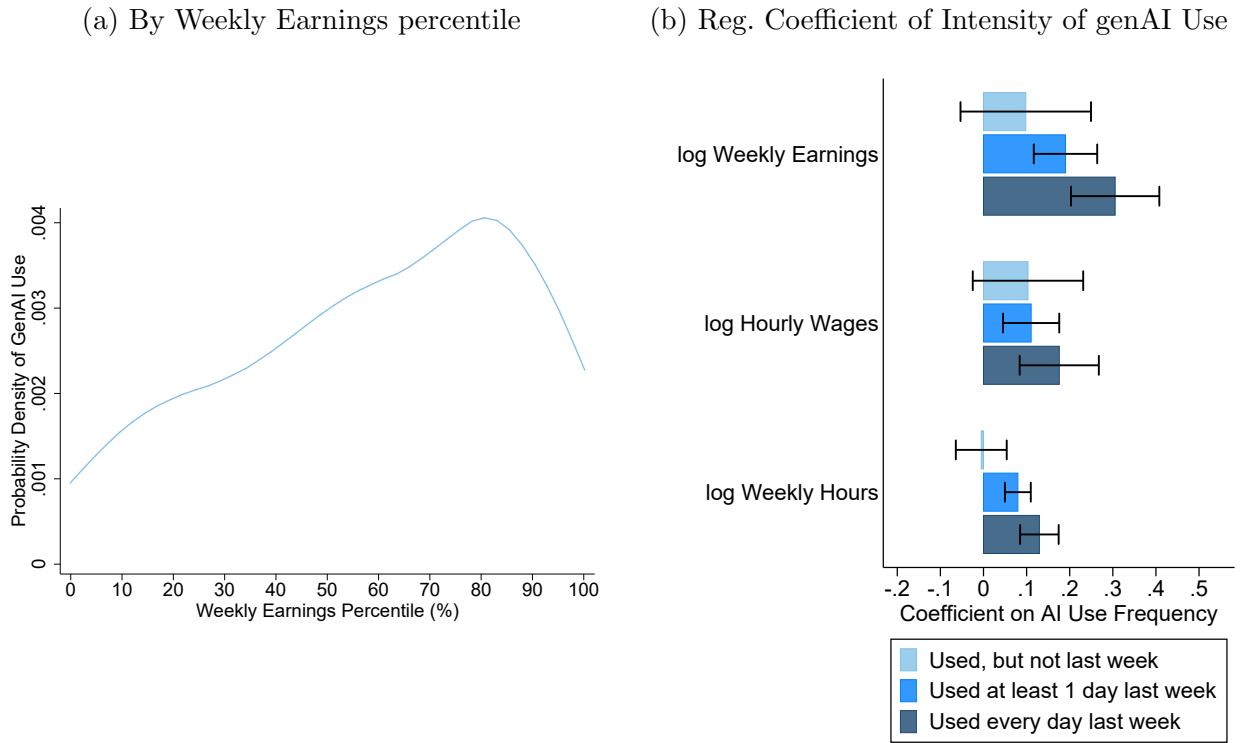


Figure 3: Demographic Heterogeneity in Work Adoption of Generative AI



Notes: The figure shows adoption rates at work for genAI by sex, age, and education. The data source is the August and November 2024 waves of the RPS. The sample is employed individuals ages 18-64 (RPS, $N = 6935$).

Figure 4: Earnings and Generative AI Adoption



Notes: Panel (a) shows adoption rates at work for genAI by weekly earnings percentile (kernel density). Panel (b) shows the coefficients on genAI intensity-of-use (zero days last week, some but not all workdays last week, and every workday last week) in a regression of (i) log weekly earnings, (ii) log hourly wages, and (iii) log weekly hours worked. For all regressions, the additional control variable used are indicators for: female, age 18-29, age 40-49, age 50-64, a bachelor's degree, a graduate degree, Black, Hispanic, occupation group, and industry. The data source for genAI is the August and November 2024 waves of the RPS. The sample is employed individuals ages 18-64 whose implied hourly wage is at least the federal minimum wage and whose weekly earnings is below \$10,000 (RPS, $N = 6063$).

rates increase steadily with income until roughly the 80th percentile, and then decline. To better understand the relationship between technology use and earnings, we regress earnings, wages, and hours on indicators of the frequency of genAI use while also controlling for occupation, industry, education, and demographic characteristics.¹¹ The results are displayed in Figure 4b. On average, genAI use is associated with a weekly earnings premium of 20 log points, a wage premium of 15 log points, and 5 log points for weekly hours. However, these average effects mask important heterogeneity. The coefficients on earnings, wages, and hours are all increasing in genAI use intensity, and infrequent users show no statistical difference between non-users.

Higher wages by (more intensive) users could reflect several factors. First, within an occupation and industry jobs that benefit more from genAI may be associated with higher wages (Deming and Kahn, 2018). Second, more productive workers may adopt new technologies more

¹¹The control variables used in these regressions are indicators for: female, age 18-29, age 40-49, age 50-54, a bachelor's degree, a graduate degree, Black, Hispanic, occupation group, and industry.

rapidly (Nelson and Phelps, 1966). Third, genAI may directly increase workers’ productivity and wages. Our data do not allow us to distinguish between these explanations.

4.1 Tasks, Predicted Exposure, and Generative AI Adoption

Why do some groups of workers adopt genAI at higher rates than others? One reason could be that workers differ in their job tasks or responsibilities, and genAI is more helpful in assisting some tasks than others.

Consistent with this explanation, we first document that genAI adoption differs greatly between occupations; see Figure 5. The three occupations with the highest genAI adoption are computer and mathematical, management, and business and finance occupations (54%, 52%, and 48%, respectively) and the bottom three are blue collar, office and administration, and personal services occupations (26%, 23%, and 16%, respectively). At the same time, genAI adoption exceeds 20% in all but one occupation groups, suggesting that the technology is useful for a broad range of occupations. As with demographic differences, differences between occupations are driven by the rates of daily and weekly genAI users.

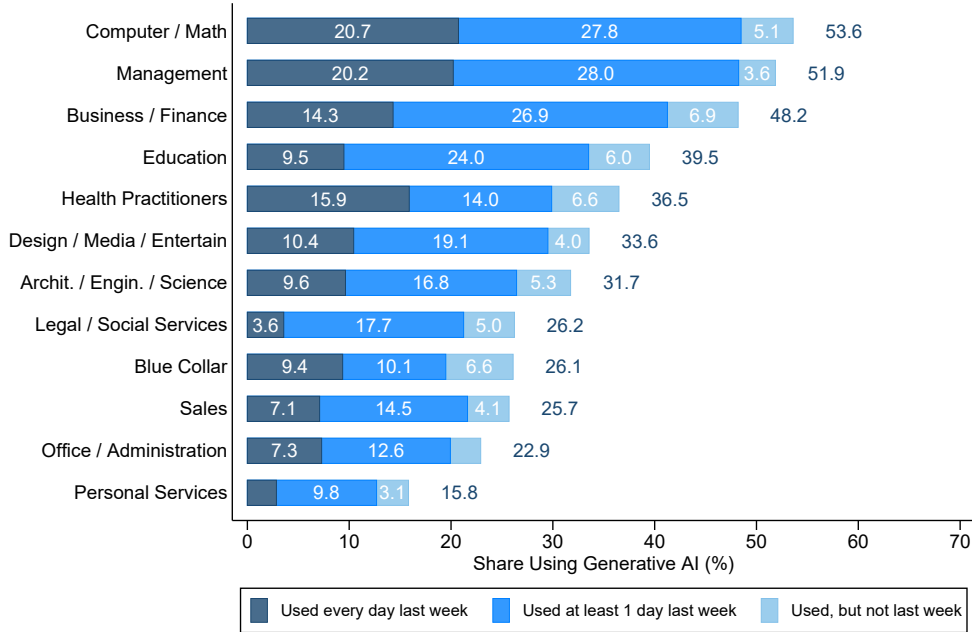
Do different adoption rates between occupations reflect differences in tasks or responsibilities? One approach to assess this is to construct task-based-exposure measures across occupations (Eloundou et al., 2024; Felten et al., 2021; Webb, 2020). For example, Eloundou et al. (2024) constructs a rubric that assigns each task in the O*NET database a zero or one corresponding to whether LLMs (a popular class of genAI) could feasibly produce large productivity gains for that task. They then construct a predicted exposure score for each occupation by aggregating task-level scores for each occupation. Using RPS data, we can ask whether these predictions are correlated with adoption. Our study is the first to evaluate these task-based “exposure” predictions using actual genAI adoption rates.¹²

Figure 6a plots an occupation group’s predicted genAI exposure from Eloundou et al. (2024) against actual genAI adoption in the 2024 RPS. We find that predicted genAI exposure is highly correlated with actual genAI adoption ($\rho = 0.65$). In Figure 6b, we find a similar correlation ($\rho = 0.71$) using the predicted exposure score from Felten et al. (2021). For example, personal service and blue-collar occupations have relatively low predicted exposure and also relatively low adoption, while computer and mathematical occupations have both high predicted exposure and high adoption.

The high correlation between predicted exposure and actual adoption provides partial validation of task-based exposure predictions, supporting their utility for researchers. At the same

¹²Handa et al. (2025) assign tasks to conversations by Claude.ai Free and Pro users and show that task usage rates are correlated with the predictions by Eloundou et al. (2024).

Figure 5: Generative AI Work Adoption By Occupation



Notes: The figure shows adoption rates at work for genAI by occupation. The data source is the August and November 2024 waves of the RPS (blue bars). The sample is employed individuals age 18-64 (RPS, $N = 6883$). We exclude workers in “agriculture / forestry / fishing” due to a very low sample size. For details on occupation categories, see Appendix A.4.

time, we also note particular occupations whose adoption rates deviate from what we would expect based on their exposure score. In particular, office and administrative occupations and managers are predicted to be similarly exposed to genAI, but actual adoption by managers is more than double that of office and administrative workers. One potential reason for these discrepancies could be that existing task-based predictions do not perfectly identify which occupations would benefit from genAI.

To investigate this possibility, we leverage additional information about workers’ adoption decisions from the RPS. Among users who did not adopt genAI, we provided a list of potential reasons why they did not adopt. We then asked respondents to select the reasons that were relevant for their decision and to rank them in order of importance. We define the share of an occupation who does not use genAI because it is not useful for their job as the share of all workers in that occupation who ranked “GenAI cannot help me with my job” in their top two reasons for not using genAI. (For the full list of potential reasons, see Appendix A.5.)

Figure 6c plots genAI adoption against the share who reported that it was not useful. We find that occupational adoption is highly correlated (-0.88) with the share who reported genAI was not useful. For example, only 7% of workers in computer / math occupations reported that genAI was not useful for their job, compared with 30% of personal service workers.